

Q1. DICTIONARY**ANSWER**

```
d1={"a":1}
d2={"b":2}
d1=update(d2)
print(d1)
output
(1,2)
if the key overlaps the d1 replace the key to d2
d=d1|d2
```

Q2. DICTIONARY**ANSWER**

shallow copy: shallow copy is used for nested objects
deep copy:
deep copy independently identifies the factors

Q3. Tuple**ANSWER**

```
t=(10,20,30,40,50)
a,b,*remaining=t
print(a)
print(b)
print(remaining)
```

Q4. DICTIONARY**ANSWER**

```
contacts={}
contacts["Ali"]="8977398980"
print(contacts ["Ali"])

output
8977398980
```

Q5. Tuple**ANSWER**

```
(1,2,3)(1,2,3)
6
```

Student 2 **Chintha Abhilash Reddy** 192525462RC

5 / 5 submitted

Q1. DICTIONARY**ANSWER**

```
dictionary is an unorded collection and which in a simple of an element in dictionary has a key: value of pair
example:
contacts = {}
contacts["ravi"] = "9182040036"
contacts["ram"] = "7670853577"
print(contacts)
name = input("enter the number")
if name in contants:
    print("phone",contacts,[name])
else:
    print("not found")
output:
{'ravi':'9182040036','ram':'7670853577'}
```

Q2. Tuple**ANSWER**

```
a.index(tuple)
ex:
a = [1,2,3,4,5]
a.index(3)
print(a)
output:
4
```

Q3. LIST**ANSWER**

```
list is one of the mostly commonly used data structures in python it is used to store multiple values in a single variable
def rotate_list(lst,k):
    k = k % len(lst)
    return list[-k:] + lst[:-k]
lst = [1,2,3,4,5]
k =2
print(rotate_list(lst,k))
output:
[4,5,1,2,3]
```

Q4. SET**ANSWER**

```
s = {1,2,3}
s.add(4)
s.remove(2)
print(s)
output:
{1,2,4}
```

Q5. Tuple**ANSWER**

```
a = (1,2,3,4,5)
a = tuple(a)
print(a)
output:
(1,2,3,4,5)
tuple are faster than list
tuple is same as list
```

Student 3 **DANIMIZA V** 192421363RC

0 / 5 submitted

Q1. DICTIONARY

— Not Submitted —

Q2. Tuple

— Not Submitted —

Q3. DICTIONARY

— Not Submitted —

Q4. SET

— Not Submitted —

Q5. DICTIONARY

— Not Submitted —

Student 4 **DEVALLA GURU RAMYA SREE** 192420061RC

5 / 5 submitted

Q1. Tuple

ANSWER

immutable- tuples value cannot change after creation
faster-tuples are faster then list because they are immutable
less memory- tuples are less memory then list
data protection

Q2. DICTIONARY

ANSWER

a dictionary stores data in the form of key : value pairs.(unique keys)
hashing is technique of cover data in a fixed size value called hash value

Q3. DICTIONARY

ANSWER

```
numbers = [1, 2, 2, 3, 3, 3, 4]
freq = {}
for n in numbers:
    freq[n] = freq.get(n,0)+1
print(freq)
ouput: {1:1, 2:2, 3:3, 4:2}
```

Q4. SET

ANSWER

```
numbers = [1, 2, 2, 3, 4, 4, 5]
numbers = list(set(numbers))
print(numbers)
output:[1, 2, 3, 4, 5]
```

Q5. Tuple

ANSWER

(10,[99,30],40)

Student 5 **Deveswar A** 192512152RC

0 / 5 submitted

Q1. Tuple— Not Submitted —

Q2. SET— Not Submitted —

Q3. LIST— Not Submitted —

Q4. SET— Not Submitted —

Q5. Tuple

— Not Submitted —

Student 6 **H Harshavardhan** 192511171RC

5 / 5 submitted

Q1. Tuple**ANSWER**

- 1.tuple is immutable which means the value cannot be changed
- 2.the tuple is faster than list
- 3.sutable for fixed data
- 4.as it was immutable the data can not be changed ,as it will have security

Q2. SET**ANSWER**

- 1.frozenset is immutable version of a set we cant add or remove elements from it
 - 2.since it is immutable , hashable ,it can also be used where an normal set cant not such as a dictionary key
 - 3.frozen set can be used as dictionary keys
 - 4.it can support the read only set operations such as union ,intersection, and membership tests
- example:
a=frozenset(['a','b','c'])
print(a)
output:
frozenset({'b','a','c'})

Q3. COPY**CODE**

```
import copy
a=[[1,2],[3,4]]
shallow=copy.copy(a)
deep=copy.deepcopy(a)
a[0][0]=100
print(shallow)
print(deep)
```

OUTPUT

(no output)

Q4. Tuple**ANSWER**

```
a=[1,2,3,4,5] //list is created
b=(1,2,3,4,5)//tuple is created
c=tuple(a) // here the list was turned into a tuple as (1,2,3,4,5), where the list() turns list into tuple
d=list(b) //here the tuple was turned into a list as [1,2,3,4,5],where the tuple() turns tuple into list
```

Q5. DICTIONARY**ANSWER**

```
d={'a':3,'b':1,'c':2}
res=dict(sorted(d.items(),key=lamda x:x[1]))
print(res)
//output
{'b':1,'c':2,'a':3}
```

Student 7 **Jayadeep J** 192511304RC

5 / 5 submitted

Q1. DICTIONARY**ANSWER**

```
d={'a':1,'b':4,'c':3}
result=dict(sorted(d.items(),key=lambda item:item[1]))
print(result)

#op: {'a':1,'c':3,'b':4}
```

Q2. LIST**ANSWER**

```
def enqueue(val):
    l.append(val)
def dequeue():
    l.pop(0)
def front():
    return l[0]
def is_empty:
    if len(l)==0:
        return True
    else:
        return False
l=[(int i) for i in input().split(" ")]
enqueue(10)
dequeue()
print(l,front(),is_empty())

#ip: 1 2 3 4 5
#op:[2,3,4,5,10] 2 False
```

Q3. SET**ANSWER**

->how data is stored:
 -->list are stored as dynamic arrays whereas sets are stored as hash tables
 -->lookup performance
 -->time complexity set:O(1) whereas for lists:O(1) to O(n)
 -->access method:
 -->sets uses a hash function to jump to the value directly whereas list scan elements sequentially
 -->insertion and deletion:
 -->sets avoid memory shifting operation whereas list require memory shifting operation

Q4. Tuple**ANSWER**

```
a=(1,2,3,4,5)
b=list(a)#here a is converted into a list then stored into b using list() function
c=tuple(b)#here b is converted into a tuple then stored into c using tuple() function
print(b,c)
#op:[1,2,3,4,5] (1,2,3,4,5)
```

->a list to tuple conversion is required when the elements should not/doesn't need to be modified
 -->whereas a tuple to list conversion is needed/used when the elements of a tuple need to be modified/removed

Q5. SET**ANSWER**

```
s1={1,2,3,4,5,6,7,8,9}
s2={6,7,8,9,10,11,12,13,14}
print("union of sets:",s1|s2)
print("intersection of sets:",s1&s2)
print("difference of sets:"s1-s2)
#op: {1,2,3,4,5,6,9,8,7,10,11,12,13,14}
      {6,8,7,9}
      {5,2,3,4,1}
#all the output will be in random order rather than the exact order it was given!!
```

Student 8 **Kampati Pavan Kumar** 192524333RC

5 / 5 submitted

Q1. SET**ANSWER**

```
A={1,2,3}
B={1,2,3,4,5}
```

SUBSET=A is said to be subset of B when all the elements in A is present in B

SUPERSET=B is said to be superset of A when all the elements in B is present in A and the number of elements in B should be greather than A

Q2. Tuple**ANSWER**

since tuple cannot be changed after creation, they reduce the possibility of accidental modification when data is shared between threads and function

Q3. Tuple**ANSWER**

```
tuple=(10,20,30,40,50)
a,b,*rest=tuple
print(a)
print(b)
print(rest)

a=10
b=20
rest=(30,40,50)
```

Q4. LIST**ANSWER**

```
lst=[1,2,3,4,5]
a=[1,2,3,4,5]
b=[1,2,3,4,6]
```

Q5. Tuple**ANSWER**

```
#tuples are immutable
#they are useful for fixed data
#they can be used as dictionary keys if their elements are hashable
#they are convinient for returning multiple values from function
```

Student 9 **KARTHIGA S** 192319057RC

5 / 5 submitted

Q1. Tuple**ANSWER**

```
def calculate(a,b):
    sum=a+b
    product=a*b
    avg=(a+b)/2

a=10
b=20
s,p,avg=calculate(a,b)

print("sum:",s)
print("average:",avg)
print("product:",p)

output :
sum:13
average:15
product: 200
```

Q2. LIST**ANSWER**

```
import math
n=5
d={i: math.factorial(i) for i in range(i,n+1)}
print(d)

output:
{1: 1 ,2: 2, 3: 6, 4: 24, 5: 120}
```

Q3. DICTIONARY**ANSWER**

```
li=list(map(int,input().split()))
freq={}
for x in li:
    freq[x]=freq.get(x,0)+1
for x , count in freq.items():
    print(x,".",count)
```

Q4. Tuple**ANSWER**

```
t1=(1,2,3)
t2=(1,2,4)
print("Tuple1:",t1)
print("Tuple2:",t2)
print("t1=t2",t==t2)
print("t1 is not equal to t2",t1!=t2)
print("t1>t2",t1>t2)
print("t1<t2",t1<t2)
```

Q5. DICTIONARY**ANSWER**

```
employees={ "Ravi" : {salary: 25,000},
            "anand":{salary:30,000},
            "aswin":{salary: 35,000} }

print(employees)
```

Student 10 **KAVI PREETHA G** 192420040RC

0 / 5 submitted

Q1. DICTIONARY

— Not Submitted —

Q2. DICTIONARY

— Not Submitted —

Q3. DICTIONARY

— Not Submitted —

Q4. DICTIONARY

— Not Submitted —

Q5. LIST

— Not Submitted —

Student 11 **Kolli Venkata Yathish Chandra** 192524323RC

4 / 5 submitted

Q1. DICTIONARY**ANSWER**

```
d={n: n**3 for n in range (1,11)}
print(d)
output: (1,8,27,64,125,260,343,512,729,1000)
```

Q2. DICTIONARY

— Not Submitted —

Q3. LIST**ANSWER**

```
lst=[1,2,3,4,2]
x=2
lst=[i for i in lst if i!=x]
print(lst)
output
[1,3,4]
```

Q4. Tuple**ANSWER**

```
print((1,2,3)<(1,2,4))
output:
Ture
```

Q5. LIST**ANSWER**

```
clear() means remove all elements but keeps same list object
del means completily delete the list variables
list[] create a new memory list makes 1st return to it
```

Student 12 **Kondam Bhaskar Tharunika** 192525276RC

5 / 5 submitted

Q1. Tuple**ANSWER**

A tuple cannot be changed after creation, so multiple threads can safely read the same tuple without worrying that another thread will modify the data this reduces race conditions and the need of lock when sharing the data only between threads

Q2. SET**ANSWER**`s={1,3,4}`**Q3. LIST****ANSWER**

```
queue = []
queue.append(10)
queue.append(20)
queue.append(30)

print("queue:"`queue)
print("deleted:",queue.pop(0))
print("front:",queue[0])
print("is empty:", len(queue) == 0)

ouput:
queue :[10,20,30]
deleted :10
front:20
is empty : false
```

Q4. DICTIONARY**ANSWER**

A Dictionary stores data in key value pairs where each key is unique and is used to access its corresponding value.
ex:{"name:" "ravi" , "age:" "22"}
Hashing: It is a process of converting a key into hash value or index using a hash function ,which helps in storing data and receive data quickly

Q5. LIST**ANSWER**

List Comprehension: It stores and creates the entire list in memory at once
Ex: Squares=[x *x for x in range(5)]
Generator: Produces one values at a time, so it uses less memory.
Ex: Squares=(x*x for x in range(5))
Difference Between These is List comprehension uses more memory ,while generator expression has efficient memory

Student 13 **Kondapalli Pujitha** 192524188RC

5 / 5 submitted

Q1. COPY**CODE**

```
import copy
original = [[1,2],[3,4]]
shallow = copy.copy(original)
deep = copy.deepcopy(original)
original[0][0] = 99
print("original:", original)
print("shallow copy:", shallow)
print("deep copy:", deep)
```

OUTPUT

(no output)

Q2. DICTIONARY**ANSWER**

```
numbers = [1,2,3,4,5,6,7,8,9,10]
result = {"even": [], "odd": []}
for num in numbers:
    if num % 2 == 0:
        result["even"].append(num)
    else:
        result["odd"].append(num)
print(result)
```

Q3. LIST**ANSWER**

clear() = Clear everything. and completely empties the list.
del = it can also delete items from list.
empty list(list = []) = there is no elements present in the list.
memory = remembers information about you.
references = sources used to support an answer.

Q4. DICTIONARY**ANSWER**

hashing is also known as searching .
in dictionary many key values are stored as;
"name" = "Rahul"
"age" = 20
"course" = "python"
student = {"name": "Rahul", "age": 20, "course": "python"}

Q5. LIST**ANSWER**

```
list = [5,10,15,20,25,30]
a)extract _ middle = list(set(list))
print("extract _ middle:", middle)
b) list = [5,10,15,20,25,30]
reversed _ list = numbers[::-1]
print(reversed _ list)
c) list =[100,200,300]
replace _ elements =list(set(list))
print("replace _ elements")
```

Student 14 **KOTA PAVITHRA** 192323011RC

5 / 5 submitted

Q1. LIST**ANSWER**

```
numbers=[1,2,3,4]
result=["Even" if n%2==0 else"Odd" for n in numbers]
print("Original:",numbers)
print("Result:",result)
output:
Original:[1,2,3,4]
Result:["Odd" ,"Even" ,"Odd" , "Even"]
```

Q2. SET**ANSWER**

```
A={1,2,3,4}
B={1,2,3,4,5}
print("A is subset of B", A.issubset(B))
print("B is superset of A", B.issuperset(A))
output:
A is subset of B:True
B is superset of A:True
```

Q3. DICTIONARY**ANSWER**

```
numbers=[1,2,3,4,5,6,7]
result={
    "Even":[n for n in numbers if n%2==0]
    "Odd":[n for n in numbers if n%2!=0]
}
print(result)
output:
Even:2, 4, 6
Odd:1,3,5,7
```

Q4. Tuple**ANSWER**

```
my_list=[10,20,30,40,50]
my_tuple=tuple(my_list)
print("List=",my_list)
print("Tuple=",my_tuple)
my_tuple2=(30,40,50,60)
my_list2=list(my_tuple2)
print("Tuple=",my_tuple2)
print("List=",my_list2)
output:
List=10,20,30,40,50
Tuple=10,20,30,40,50
Tuple=30,40,50,60
List=30,40,50,60
```

Q5. DICTIONARY**ANSWER**

```
employees={
    "Pavithra":2000
    "kavitha":5000
    "preethi":6000
}
print(employees["kavitha"])
output:
5000
```

Student 15 **Madireddy Gowtham Kumar Reddy** 192572066RC

4 / 5 submitted

Q1. DICTIONARY

ANSWER

```
employees = {  
    "E101" : {"name":"Gowtham","salary":30000}  
    "E102" : {"name":"Rehan","salary":35000}  
    "E103" : {"name":"Praneeth","salary":40000}  
}  
print("salary of E102 : ",employees["E102"]["salary"])
```

output:
salary of E102 : 35000

Q2. DICTIONARY

ANSWER

python dictionary store data as key value pairs inside underlying hash tables instead of a sequence index keys are mapped specific memory location allowing average time complexit of $O(1)$.
hashing is a process of key thought mathematical algorithm to converted it into a fixed size integer is known as a hashing.

Q3. LIST

ANSWER

```
q = []  
q.append(10)  
q.append(20)  
print("queue:",q)  
print("dequeue:",q.pop[0])  
print(len(q[0]))
```

Q4. DICTIONARY

ANSWER

```
number = [1,2,3,4,5,6,7,8,9,10]  
result = {  
    "even" : [n for n in number if n%2 == 0]  
    "odd" : [n for n in number if n%2! == 0]  
}  
print(result)
```

Q5. LIST

— Not Submitted —

Student 16 **MALAVIKAH V** 192419061RC

5 / 5 submitted

Q1. LIST**ANSWER**

```
def rotate_list(lst,k):  
    return lst[-k:] + lst[:-k]  
lst=[1,2,3,4,5]  
k=2  
print(rotate_list(lst,k))  
  
output:  
[4,5,1,2,3]
```

Q2. DICTIONARY**ANSWER**

```
output:  
{'x':1, 'y':2, 'z':3}
```

Q3. LIST**ANSWER**

```
student={  
    "name":"rahul",  
    "sub": ["python", "dsp"],  
    "marks": (85,90),  
    "club": {"sports,music"}  
}  
  
dictionary - stores student details  
list - stores sub  
tuple - stores marks  
set - stores club
```

Q4. DICTIONARY**ANSWER**

```
d={n:n**3 for n in range(1,11)}  
print(d)  
  
output:  
{1:1,2:8,3:27,4:64,5:125,6:216,7:343,8:512,9:729,10:1000}
```

Q5. Tuple**ANSWER**

```
t=(10,20,30,40,50)  
a,b,*c=t  
print(a)  
print(b)  
print(c)  
  
output:  
10 20 [30,40,50]
```

Student 17 **Mohammed Subahaan H** 192572165RC

5 / 5 submitted

Q1. Tuple

ANSWER

```
a=[1,2,3,4]          #list is created
t=(4,5,6,7)          #tuple is created
b=tuple(a)            #tuple( ) will convert list to tuple
c=list(t)             #list( ) will convert tuple to list
```

Q2. LIST

ANSWER

```
-> python lists are mutable
-> ordered collection of data
-> list elements are enclosed within [ ]
-> dynamic resizing means size grows and slicing can be done
-> heterogeneity means it can store different type of data together
eg: a=[1,"List",2.03,"chennai"]
```

Q3. Tuple

ANSWER

```
(1,2,3,1,2,3)
6
```

Q4. LIST

ANSWER

```
[1,2,3,4][1,2,3,4,5][1,2,3,4,6]
```

Q5. SET

ANSWER

```
-> set is a unordered collection of elements enclosed within curly braces{ }
-> mutable
-> Does'nt support indexing
-> it stores only unique values , if duplicates are added , the python ignores it.
eg : a={1,2,3,4}
```

Student 18 **Moniga V** 192521184RC

5 / 5 submitted

Q1. SET**ANSWER**

```
num = [1,2,3,2,4,1,5,3]
unique_numbers = list(set(numbers))
print("Original list: ", numbers)
print("List without duplicates: ", unique_numbers)
```

Q2. SET**ANSWER**

```
my_set = frozenset([1,2,3])
print(my_set)
```

If we try to add something we will get error because frozenset cannot be modified.
It is immutable because we can't
add elements
remove elements and
change existing elements

Q3. LIST**ANSWER**

```
[1,2,3,4,5,6] [1,2,3,4,5,6] [1,2,3,4]
```

Q4. DICTIONARY**ANSWER**

```
d = {n:n**3 for n in range(1,11)}
print(d)
```

```
{1: 1, 2: 8, 3: 27, 4: 64, 5: 125, 6: 216, 7: 343, 8: 512, 9: 729, 10: 1000}
```

Q5. DICTIONARY**ANSWER**

```
d = {'a': 3, 'b': 1, 'c': 2}
sorted_d = dict(sorted(d.items(), key = lambda x: x[1]))
print(sorted_d)
```

```
{'b': 1, 'c': 2, 'a': 3}
```

Student 19 **Mrudhula K** 192524335RC

5 / 5 submitted

Q1. Tuple

ANSWER

```
(1,2,3,1,2,3)
6
```

Q2. LIST

ANSWER

```
fact=1
n=5
for i in range(1,n+1):
    fact=fact*i
    i+=1
d={n : fact}
print(d)
output={5 : 120}
```

Q3. DICTIONARY

ANSWER

```
d1={"ram":1,"ravi":2}
d2={"rani":3,"raj":4}
d1.update(d2)
print(d1)
output={"ram":1,"ravi":2,"rani":3,"raj":4}
if the keys overlap in dictionary they take the first first value that assigned to them
for example if key 1 takes value ram in d1 and raj in d2, it automatically assigns value ram to the key 1
```

Q4. COPY

CODE

```
import copy
a=[[1,2],[3,4]]
b=copy.copy(a)
c=copy.deepcopy(a)
a[0][0]=2
a[0][1]=3
print(b)
print(c)
```

OUTPUT

(no output)

Q5. SET

ANSWER

no sets are not always faster than lists.
we can use sets if we need all the distinct elements, here list is unordered whereas set is ordered or unordered sometimes.
there are only few functions in set whereas list has so many functions so it might not be faster than lists in some cases.
for example sets don't support indexing but in list it supports indexing so searching an element using list is faster and easier than sets.

Student 20 **Nichenametla Venkata Mani Chandan** 192525392RC

5 / 5 submitted

Q1. LIST**ANSWER**

```
students = [{"S01","pavan",{"python","math"}},{"S02","Ravi",{"python","DSA"}}]
marks = {"S01":85,"S02":90}
```

Q2. SET**ANSWER**

Immutable means when the set is created that's can't change is known as immutable.
FROZENSET: A frozenset is an immutable version of a set. Once created its elements cannot be added or removed is known as frozenset.
Example:({1,2,3})

Q3. LIST**ANSWER**

```
a = [1,2,3,2,,4,2]
x = 2
a = [i for i in a if i != x]
print(a)
OUTPUT:
[1,3,4]
```

Q4. LIST**ANSWER**

```
queue = []
queue.append(10)
queue.append(20)
queue.append(30)
print(queue.pop[0])
print(queue[0])
print(len(queue) == 0)
OUTPUT:
10
20
False
```

Q5. SET**ANSWER**

```
A={1,2,3}
B={1,2,3,4,5}
```

SUBSET = A is said to be subset of B when all the elements in A is present in B

SUPERSET = B is said to be superset of A when all elements in B is present in A and the number of elements in B should be greater than A

Student 21 **Nithish S** 192521249RC

5 / 5 submitted

Q1. LIST**ANSWER**

Rotate a list by K positons means moving the last k elements to the begainig of the list
ex:
[1,2,3,4,5,6]
k=2
[6,5,4,3,2,1]

Q2. Tuple**ANSWER**

```
a=(10,20,10,30,10)
b=10
print("occurences:",t.count(b))
print("index:",t.index(b))
this is the code
t.count(x)=it will counts how many times b has occurs in the tuple
t.index(x)=it five the positon of the first occurence of b
```

Q3. SET**ANSWER**

The set is collection of unique elements.
Duplicates are not allowed because a set is used to store the distinct values

Q4. DICTIONARY**ANSWER**

it stores the persons name and phone number
nithish=name(key)
8838226971=value

Q5. LIST**ANSWER**

```
a=print(lst[2:5])
b=print(lst[:4]:-1)
c=lst[-2]=(100,200,300)
print(lst)
```

Student 22 **P V Karishmika** 192511453RC

0 / 5 submitted

Q1. LIST

— Not Submitted —

Q2. Tuple

— Not Submitted —

Q3. DICTIONARY

— Not Submitted —

Q4. DICTIONARY

— Not Submitted —

Q5. DICTIONARY

— Not Submitted —

Student 23 **PATAN MOHAMMED RIYAZ KHAN** 192572126RC

5 / 5 submitted

Q1. DICTIONARY**ANSWER**

Generally dictionary used key values for searching the elements Whereas List uses the integers in square brackets [] to search the elements. Dictionaries are faster for searching an element then List.

Q2. SET**ANSWER**

SET IS A MAIN CONCEPT IN PYTHON USED TO READ AND WRITE A GROUP OF ELEMENTS IN A SET. THE MAIN PROPERTIES ARE UNION, INTERSECTION, DIFFERENCE, DISJOINT & JOINT. SET IS IMMUTABLE. GENERALLY SET IS DENOTED BY "set()". WE KNOW THAT SETS DO NOT ALLOW DUPLICATE ELEMENTS BECAUSE SETS REPRESENTS EACH ELEMENT ONLY ONCE AT A TIME. SET AUTOMATICALLY SORT THE ELEMENTS IN ASCENDING ORDER. IT ALWAYS REMOVE THE DUPLICATE ELEMENTS.

Q3. Tuple**ANSWER**

```
n=(1,2,3,4,5)
a, b,*c=n
print(a)
print(b)
print(c)
```

OUTPUT:
a=1
b=2
c=[3,4,5]

Q4. DICTIONARY**ANSWER**

Generally we use dictionary for accessing elements using key values. But this is not possible in list or set. We can access the list or set are access only by using square brackets []. So we use dictionary to access the same function repeatedly using key values which was assigned by us which was not possible in set or list.

Q5. DICTIONARY**ANSWER**

Dictionary is also known as hash map. It is mutable. It stores data strictly in the Key-value pairs. Generally it stores key values in the curly braces { }. Hashing is the process of converting an arbitrary size input into a fixed size.

Student 24 **PILLARI RAMYA** 192319027RC

5 / 5 submitted

Q1. Tuple**ANSWER**

```
def calculate(a,b):
    sum = a+b
    product=a*b
    avg = (a+b)//2
a= 10
b = 20
s,p,avg = calculate(a,b)
print("sum=",s)
print("product=",p)
print("average=", "avg")
output:
sum=30
product=200
average= 15
```

Q2. SET**ANSWER**

In a set discard() and remove() methods will be present. Discard(x) - means it will safely remove "x" and if "x" is not in function ,it does not do anything. Remove(x)- means it will also removes the "x" but if "x" is not present it crashes the entire program.

Q3. LIST**ANSWER**

```
[1,2,3,4,5]
[1,2,3,4,5]
[1,2,3,4,6]
```

Q4. LIST**ANSWER**

```
q = [ ]
q.append(10)
q.append(20)
print(q.pop(0))
print(q[0])
print(len(q)==0)
output:
10
20
false
```

Q5. LIST**ANSWER**

```
lst = [5,10,15,20,25,30]
print(lst[2:5])
lst[:4] = lst[:4][::-1]
print(lst)
lst[-2:] = [100,200,300]
print(lst)
output:
[15,20,25]
[20,15,10,5,25,30]
[20,15,10,5,100,200,300]
```

Student 25 **POTHIREDDY ROHINI** 192420054RC

5 / 5 submitted

Q1. DICTIONARY**ANSWER**

A DICTIONARY IS USED WHEN IT NEEDS TO STORE A KEY VALUES PAIR AND ACCESS USING UNIQUE KEY.
PS:STUDENT RECORD
EMPLOY DETAILS

Q2. DICTIONARY**ANSWER**

```
d={"name": "rohini", "age":20}
print(d.get("name"))
print(d.get("marks", "not found"))
output:
rohini
20
not found
```

Q3. LIST**ANSWER**

```
squares =[x**2 for x in range(1,21)if x%2 !=0]
print(squares)
output:[1,9,25,49,81,121,169,225,289,361]
```

Q4. LIST**ANSWER**

hashability means an object has a fixed hash value and that can be use as an element in a set.
list cannot be element of set because they are mutable.
tupules can be elements of set because they are immutable

Q5. Tuple**ANSWER**

output:3

Student 26 **Risaph S** 192519096RC

5 / 5 submitted

Q1. DICTIONARY**ANSWER** `{'x': 1, 'y': 2, 'z': 3}`**Q2. DICTIONARY****ANSWER** the dictionary will be {0:0,1:1,2:4} . the sum of values will be 0+1+4 then the output will be 5 so the output is 5**Q3. SET****ANSWER** Subset: All of its elements is contained by other sets
Superset: It is type of set which contain the elements of other sets**Q4. SET****ANSWER** `A={1,2,3,4}`
`B={3,4,5,6}`
`print(A|B)`
`print(A&B)`
`print(A-B)`
output:
`{1,2,3,4,5,6}`
`{3,4}`
`{1,2}`**Q5. Tuple****ANSWER** tuple is an immutable one because its elements cannot be changed or modified but if it can contain mutable objects such as list which can be modified and changed
for example:
`t = (20,[30,40],50)`
`t[1][0]=87`
`print(t)`
output:
`(20,[87,40],50)`

Student 27 **Sachin S** 192525045RC

5 / 5 submitted

Q1. DICTIONARY

ANSWER

```
contacts={}
contacts["sachin"] = "8825417607"
contacts["risaph"] = "6369714577"
contacts["smohit"] = " 8637471946"
print("contact book:")
for name, number in contact.item():
    print(name, ":", number)
```

OUTPUT:

```
contact book:
sachin = 8825417607
risaph = 6369714577
smohit =8637471946
```

Q2. LIST

ANSWER

List comprehension will create complete list and store all element in memory and the generator expression use to produce one element at a time and the memory usage is less

Q3. DICTIONARY

ANSWER

```
{'x':1,'y':2,'z':3}
```

Q4. DICTIONARY

ANSWER

5

Q5. DICTIONARY

ANSWER

count

Student 28 **SAFRIN RITHA AM** 192313059RC

5 / 5 submitted

Q1. SET**ANSWER**

discard()-safely delete's 'x'.if 'x' does not delete x exists then it gracefully does nothing instead of crashing
remove()-it delete's 'x' from the set .if the 'x' does not exist then it will crashes the program

Q2. DICTIONARY**ANSWER**

we use dictionary instead of list or set because it mutable collection are store the data in key-value pairs only.
user={ "name:" "safrin"
"age:" "20"}

Q3. SET**ANSWER**

```
s={1,2,3}
s.add(4)
s.remove(2)
print(s)
output:{1,4,3}
```

Q4. Tuple**ANSWER**

```
tuple=(1,2,3,4,5)
a,b,*rest=tuple
print("a =",a)
print("b =",b)
print("remaining values:",rest)
outputs: a=1
b = 2
remining values:3 4 5
```

Q5. SET**ANSWER**

```
a = [1,2,2,3,3,4]
b=list(set(a))
print(b)
```

Student 29 **Shaik Rehan** 192525110RC

4 / 5 submitted

Q1. LIST**ANSWER**

```
squares = [x**2 for x in range(1,21,2)]  
print(squares)
```

```
output:-  
(1,9,25,49,81,121,169,225,289,361)
```

Q2. Tuple**ANSWER**

```
my_tuples = [10,20,30,20,40,20]  
occurrences = my_tuple.count(20)  
first_index = my_tuples.index(20)  
print(occurrences)  
print(first_index)
```

```
output:-  
3  
1
```

Q3. DICTIONARY

— Not Submitted —

Q4. LIST**ANSWER**

```
lst = [1,2,3,4]  
a = lst  
b = lst  
a.append(5)  
b.append(6)  
print("Lst:",lst)  
print("a:",a)  
print("b:",b)
```

```
output:-  
lst:[1,2,3,4]  
a:[1,2,3,4,5]  
b:[1,2,3,4,6]
```

Q5. COPY**CODE**

```
import copy  
original = [[10,20],[30,40]]  
shallow = copy.copy(original)  
deep = copy.deepcopy(original)  
original[0][0]=99  
print(shallow[0][0])  
print(deep[0][0])
```

OUTPUT

(no output)

Student 30 **SIVACHANDRABOSE S** 192319030RC

4 / 5 submitted

Q1. DICTIONARY

— Not Submitted —

Q2. LIST**ANSWER**

```
a=[10, 20, 30]
b=a
a.clear()
print(a)
print(b)
```

Q3. Tuple**ANSWER**

```
t=(10, 20, 30, 40, 50)
a,*b =t
print("First value:", a)
print("Remaining values:", b)
```

Q4. LIST**ANSWER**

A list is an ordered and mutable collection used to store multiple values in single variable.

Q5. SET**ANSWER**

A set represent unique elements. Repeating an element does not provide any new information.

Student 31 **Smohit Arman Swain** 192572092RC

5 / 5 submitted

Q1. DICTIONARY

ANSWER

USE DICTIONARY WHEN WE NEED FAST AND QUICK RESPONSE.USE A DICTIONARY WHEN A DATA HAS A KEY OR VALUE.

SETS ONLY STORES UNIQUES DATA WHEREAS DICTIONARY CAN STORE DIFFERENT DATAS OF SAME TYPES

PRACTICAL SCENARIOS:

1. SCHOOL RECORD
2. OFFICE PERSONAL DETAILS RECORDS

Q2. SET

ANSWER

it is a immutable set as we cant add or remove elements. frozenset is a set when once created elements cant be added or removed from set and hence also immutable

example: `F = FROZENSET([1,2,3])`

```
print(f)
OUTPUT= [1,2,3]
```

Q3. LIST

ANSWER

we can remove all occurrences of a given element from a list without using `remove()` repeatedly by using `filter()`

Q4. LIST

ANSWER

list comprehension creates list , it is more fast when compared to generator ,it stores all its data in the memories, it use `[]` brackets

whereas

Generator expressions generates object,it is much more efficient when compared to list,it does not store all the data in its memory , it uses `()`

Q5. LIST

ANSWER

hashability in a set is a fixed value that does not change for its lifetime.it is made so that we dont misclick and change the value and also it is also used to find a value which we dont know where it is in a large pile of datas , we can simply search for this fixed value as it doesnt change.

list cant be elements of a set as list is mutable and set is immutable hence it cant be elements of sets

Student 32 **SUNDARA VARADHAN K S** 192319033RC

5 / 5 submitted

Q1. Tuple

ANSWER

```
t=(10,20,30,40)
a, b *rest=t
print(a, b, rest)
```

Q2. SET

ANSWER

```
a=[1, 2, 3, 4]
b=[3, 4, 5, 6]
common=set(a) and set(b)
print(common)
```

Q3. LIST

ANSWER

```
list = [1, 2, 3, 2, 4, 2]
x= 2
list = [ i for i in list if i = x]
print(list)
```

Q4. LIST

ANSWER

```
squares = [x**2 for x in range (1,21,2)]
print(squares)
```

Q5. Tuple

ANSWER

```
my_list = [1, 2, 3]
my_tuple = tuple(my_list)
new_list = list(my_tuple)
print(my_tuple)
print(new_tuple)
```

Student 33 **Thanikonda Aditya** 192565068RC

5 / 5 submitted

Q1. SET

ANSWER

```
a=[1,2,3]
b=[1,4,5]
common=a&b
print(common)
```

Q2. DICTIONARY

ANSWER

dictionary is also known as hash map
dictionaries stores the key values in curly braces{ }
hashing the process of converting an input of arbitrary size into fixed size

Q3. DICTIONARY

ANSWER

```
num=[1 2 3 4 5 3 4 5 6 6 3 2]
frequency={}
for num in numbers:
    if i in frequency:
        frequency[num]+=1
    else:
        frequency[num]=1
print(frequency)
```

Q4. LIST

ANSWER

```
students=[
    {
        "name": "raj"
        "age":18,
        "subjects":(
            "python",
            "java",
            "english"
        )
    }
]
unique_subject=set( )

students=[
    {
        "name": "teju"
        "age":19
        "subjects":(
            "python",
            "java",
            "maths"
        )
    }
]
unique_subject=set( )
```

Q5. DICTIONARY

ANSWER

```
{'x':1,'y':2,'z':3}
```

Student 34 **Thumati Sai Praneeth** 192525434RC

5 / 5 submitted

Q1. DICTIONARY**ANSWER**

```
num = {1,2,3,4,5,6,7,8,9,10}
grouped_numbers = {"even":[],"odd":[]}
for i in numbers:
    if i%2==0:
        grouped_numbers["even"].append(num)
    else:
        grouped_numbers["odd"].append(num)
print(grouped_numbers)
even = {2,4,6,8,10} odd = {1,3,5,7,9}
```

Q2. DICTIONARY**ANSWER**

dictionaries are faster than lists
list - $O(1)$ and $O(1)$ dictionaries - $O(1)$ and $O(n)$

Q3. DICTIONARY**ANSWER**

dictionaries are mutable the data stored as keys and values they are assigned in {}
hasing is a process converting key value into fixed numerical value
ex: d={"name": "praneeth", "age": 20, "subject": "python"}
from above code keys are (name, age, subject)
values (praneeth, 20, python)

Q4. LIST**ANSWER**

```
list = [5,10,15,20,25,30]
print(list[1:4])
print(list[::-1])
print(list[1:-1])
15,20,25
30,25,20,15,10,5
10,15,20,25,30
```

Q5. SET**ANSWER**

frozenset is version of set once created its element cannot be added or removed is known as frozen set
immutable means cannot change the assigned values once set is created so, frozenset is immutable
ex: {(2,3,4)}